Benjamin David Gibson

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EDUCATION

University of Pittsburgh — Pittsburgh, PA
B.S. in Mechanical Engineering — Expected May 2029
Relevant Coursework: Statics, Materials Science, Engineering Design, Calculus 3, Linear Algebra

The Pennington School | High School Diploma — Graduated June 2025

PROJECTS

Rover Wheel Design | Pitt Engineering Design Course — *SolidWorks • 3D Printing*

- Designed a rover wheel in SolidWorks with grouser tread geometry to improve traction.
- Compared PLA, PETG, and TPU and evaluated print orientation to select a material and manufacturing approach.
- Refined the axle-hole geometry for a friction fit, eliminating the need for hot glue in assembly.

Alexa + Claude AI Voice Assistant | Independent Project — *AWS Lambda • Claude API*

- Built an Alexa skill that sends voice requests through AWS Lambda to Anthropic's Claude API, turning an Echo Dot into a conversational assistant.
- Configured API authentication and deployed the code through the Amazon Developer Portal and AWS.
- Debugged and modified AI-generated code to resolve integration and runtime errors, resulting in a functional system.

More projects in my portfolio: ben.gibson.io
Includes a Python/SQLite market-data tracker and my SolidWorks designs. Scan the QR code to view.

TECHNICAL EXPERIENCE

Team Toyota of Princeton — Princeton, NJ
Lube Technician — May 2025 – Present

- Perform high-volume express maintenance, including oil changes, tire rotations, and multipoint inspections.
- Identify additional service needs and complete repairs such as tire replacements, wiper and air filter installations, and lighting replacements.

ADDITIONAL EXPERIENCE

Soulberry Natural Market — New Hope, PA
Juice Bar Associate — June 2021 – May 2025

- Prepared smoothies and juices to order while providing knowledgeable customer service in a fast-paced environment.
- Cross-trained across deli, cashier, grocery, and float positions to build a broad understanding of store operations.
- Proposed and implemented a shelf-divider system to fix back-storage and sales-floor inefficiencies; it remained in use after I left.

TECHNICAL SKILLS

Programming & Platforms: Java, C, Python, AWS (Lambda)
Engineering Software: MATLAB, SolidWorks
Mechanical: Automotive Maintenance, Automotive Repair, 3D Printing

ADDITIONAL

Presentation: Pitt First Year Engineering Conference – 3D Printing in Automotive Engineering
Certifications: Toyota Maintenance Certified
Languages: English (Fluent), French (Proficient)
Interests: Automotive, Engineering, Guitar, Piano